

Counterfactual Forks Teacher Notes

TOK Cognitive Science Activity Bank • History • 30-40 min

Category	History	Time	30-40 min
Difficulty	Medium	ID	counterfactual-forks

Big idea: Counterfactual thinking can clarify causation when it is disciplined by evidence.

Overview

Students build careful counterfactuals to test causal claims without turning history into fantasy.

Student Prompt

Inspect Counterfactual Forks Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Counterfactual Forks Stimulus A	Student-facing stimulus 1: School example: What if the viral incident had not happened, but earlier complaints still existed?. Mark what the source shows, what it omits, whose perspective appears, and what claim it can responsibly support.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Counterfactual Forks Stimulus B	Student-facing stimulus 2: Historical prompt: What if one reform law had failed, while economic and social pressures remained?. Mark what the source shows, what it omits, whose perspective appears, and what claim it can responsibly support.	Use this for comparison, a second round, or a partner challenge.
Counterfactual Forks Reveal / Twist	Student-facing stimulus 3: Historical prompt: What if one reform law had failed, while economic and social pressures remained?. Mark what the source shows, what it omits, whose perspective appears, and what claim it can responsibly support.	Teacher reveal: connect the changed response to the big idea — Counterfactual thinking can clarify causation when it is disciplined by evidence.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Counterfactual Forks Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Counterfactual Forks Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	Can historians know what did not happen?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how counterfactual thinking can clarify causation when it is disciplined by evidence.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Choose a causal story where one small change is plausible.
- Keep students close to evidence by requiring a constraint card for every counterfactual.

Ready-to-Use Examples

- School example: What if the viral incident had not happened, but earlier complaints still existed?
- Historical prompt: What if one reform law had failed, while economic and social pressures remained?

Suggested Timing

Stage	Teacher move	Watch for
1	Choose one historical event or school-history scenario with several causes.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students identify one plausible turning point.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups change one condition while keeping other constraints realistic.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students predict what would likely, possibly, and unlikely change.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief how counterfactuals can test causal claims.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- A disciplined counterfactual changes one thing and respects the rest of the evidence.
- What does your fork reveal about causal importance?

Common Misconceptions

- Students may treat sources as transparent windows rather than selected traces.
- Students may confuse explaining context with excusing actions.

Assessment Look-Fors

- Uses source evidence while naming selection, silence, and perspective.
- Distinguishes historical claim, interpretation, and judgement.

Differentiation and Adaptations

- Support: provide the changed condition.
- Challenge: require likely, possible, and unlikely outcomes.

Extension Tasks

- Students add counterfactual notes to an existing cause web.
- Compare counterfactual reasoning in history with model testing in science.

Related Activities

- Cause Web
- Timeline Compression Trap
- Presentism Trial